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Started	Tuesday, 20 October 2026, 6:47 PM
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Duration	3 mins 53 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n ✓
c. n-1
d. log n

Question **2**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search does not require dividing the list
d. Binary search works on unsorted lists

Question **3**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. In the right sublist ✓
c. At the beginning of the list
d. In the middle of the list

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list is unsorted
c. When the list is sorted ✓
d. When the list contains duplicate elements

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array
b. There must be a mechanism to access middle element directly
c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **6**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Sort the list ✗
b. Divide the list into two equal parts
c. Compare the target element with the middle element in the list
d. Compare the target element with the first element in the list

Question **7**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O(log n)
c. O(1) ✓
d. O👎

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Linear search ✓
- b. Quick search
- c. Merge search
- d. Bubble search

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 99 ✓
- b. 94 and 99
- c. 89 and 94
- d. 90 and 100

Question **10**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Binary search
- b. Hashing
- c. Linear search ✓
- d. Jump search

Question **11**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The list is sorted
- b. The search continues in the right sublist
- c. The search continues in the left sublist
- d. The search ends successfully ✓

Question **12**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Selection

c. Complexity

d. Bubble ✓

Question **13**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Hash

b. Both (1) & (3)

c. Linear

d. Binary ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing

b. Dynamic search

c. Binary search

d. Linear search ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Deleting elements from a list
- c. Inserting elements into a list
- d. Sorting elements in a list